

# CARDIO:DE++

Powering Next-Generation Cardiovascular AI

with a Rich, Multi-Modal German Clinical Corpus

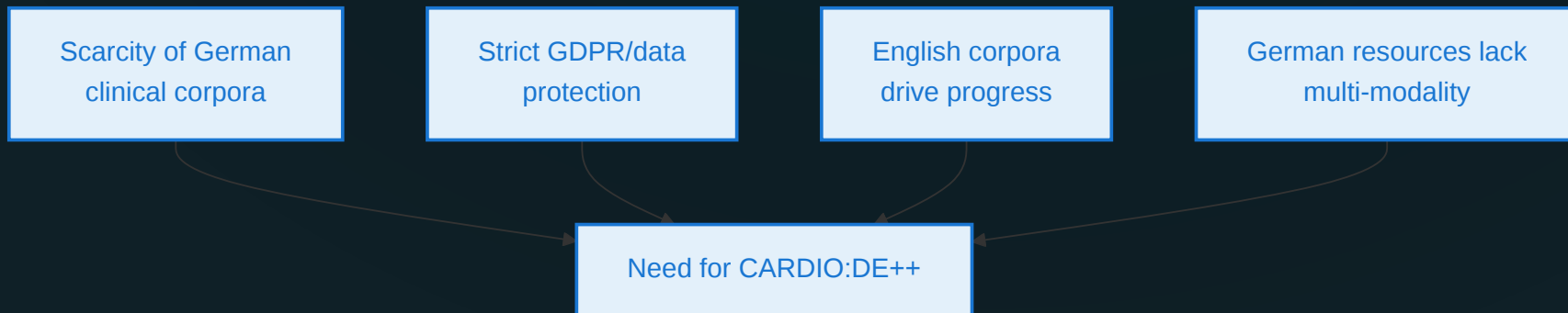


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# Motivation

- Scarcity of high-quality, freely accessible German clinical text corpora 
- Strict data protection (GDPR) limits data sharing 
- Existing English corpora (e.g., MIMIC-III, i2b2) have driven clinical NLP 
- German resources are limited and lack multi-modal, parallel data 



# State of the Art

- **English corpora:** MIMIC-III, MIMIC-IV-Note, i2b2, CLEF, SemEval, THYME
- **European corpora:** MERLOT (French), IULA (Spanish)
- **German corpora:** GGPONC 2.0, 200 Oncological Discharge Summaries, GRASSCO, JSynCC
- **Gap:** No large, multi-modal, parallel German cardiovascular corpus

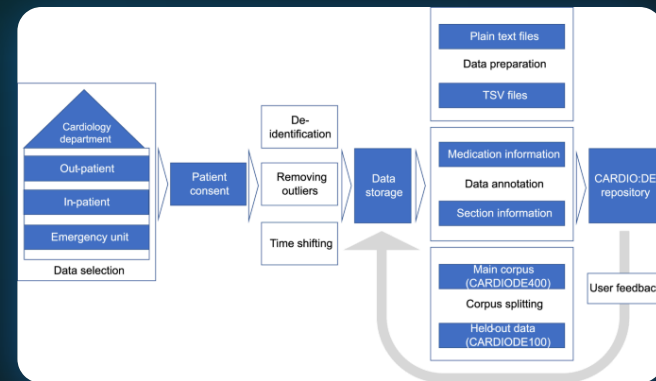


*MIMIC-III: A leading English clinical corpus.*

*Source: PhysioNet*

# Preliminary Work

- Developed de-identification and NER methods for German medical texts
- Achieved F2-scores up to 96% for de-identification
- Created and published the CARDIO:DE corpus:
  - 500 cardiovascular doctor's letters
  - 993,143 tokens, 31,952 unique tokens
  - High-quality medication and section annotations
- CARDIO:DE is freely available for research



*Figure 3: CARDIO:DE annotation statistics and inter-annotator agreement.*

*Source: Richter-Pechanski et al.,  
Scientific Data 2023*

# Why CARDIO:DE++?

- Existing corpora lack: 
- Multi-site data (risk of site bias) 
- Explicitly parallel, structured tabular data 
- CARDIO:DE++ will: 
- Add data from Universitätsklinikum Mannheim 
- Link free-text with structured clinical parameters 
- Enable advanced text-to-data and data-to-text AI models 

# Project Objectives

- **Expand corpus heterogeneity:**

Add 500 new de-identified discharge letters from two sites

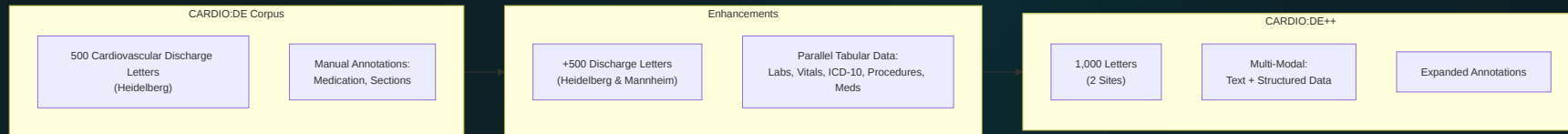
- **Integrate structured tabular data:**

Laboratory values, vital signs, ICD-10 codes, procedures, medications

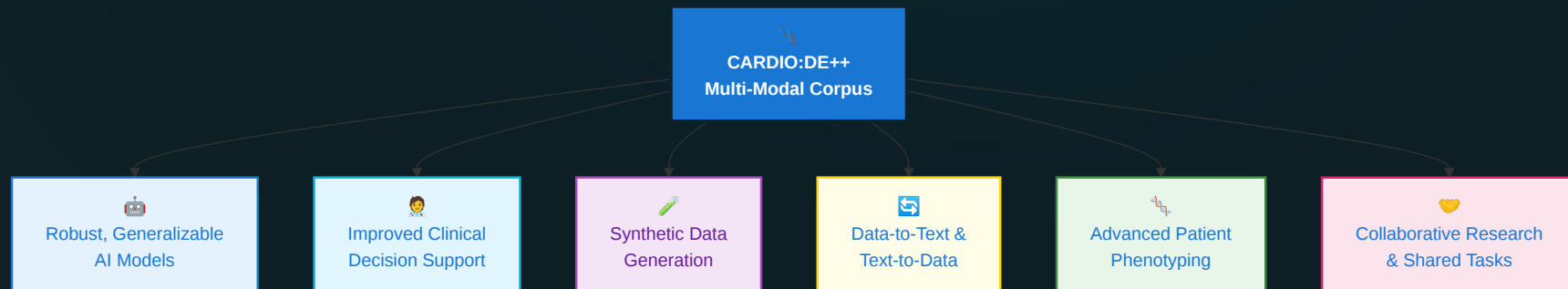
- **Enable advanced AI research:**

- Synthetic data generation
- Data-to-text and text-to-data NLP
- Patient phenotyping and subgroup discovery

# CARDIO:DE++ Enhancements



# CARDIO:DE++ Enablements



# Work Packages Overview

| WP  | Title                                | Months |
|-----|--------------------------------------|--------|
| WP1 | Ethics, Legal Framework & Onboarding | 1–3    |
| WP2 | Data Acquisition                     | 1–12   |
| WP3 | De-identification                    | 2–12   |
| WP4 | Annotation                           | 6–24   |
| WP5 | Advanced AI Model Development        | 19–23  |
| WP6 | Dissemination & Sustainability       | 19–24  |



# WP1: Ethics, Legal Framework & Site Onboarding

- **Months:** 1–3
- **Objective:** Secure ethical/legal approvals and establish data sharing protocols
- **Key Tasks:**
  - Ethics applications for both sites
  - Standardized patient consent forms
  - Data Use/Transfer Agreements (DUAs/DTAs)
  - SOPs for data handling and training
- **Deliverables:** Approved ethics, signed DUAs/DTAs, consent forms, SOPs
- **Lead:** Heidelberg (with UMM)



## Further reading:

- [GDPR \(General Data Protection Regulation\)](#)

## WP2: Data Acquisition

- **Months:** 1–12
- **Objective:** Collect 500 new discharge letters and structured tabular data
- **Key Tasks:**
  - Patient consent and data collection at both sites
  - Finalize list of structured data points (labs, ICD-10, vitals, meds, procedures)
  - Extract and link data for each letter
- **Deliverables:** 500 raw discharge letters + structured data
- **Lead:** Heidelberg & UMM



**Reference:** Johnson AEW et al., MIMIC-III Data Collection Protocol, Scientific Data (2016)

## WP3: De-identification

- **Months:** 2–12
- **Objective:** De-identify all textual and tabular data
- **Key Tasks:**
  - Apply/refine CARDIO:DE de-identification models
  - Manual review and correction
  - Clean, standardize, and anonymize tabular data
- **Deliverables:** 500 de-identified letters, linked tabular data, quality report
- **Lead:** Heidelberg



**Key resource:** Meystre et al., Review of De-identification Techniques for Clinical Text, JAMIA (2010)

## WP4: Annotation

- **Months:** 6–24
- **Objective:** Annotate new data using established schemes
- **Key Tasks:**
  - Apply CARDIO:DE annotation schemes (medication, section classes)
  - Inter-annotator agreement studies
  - Link text to tabular data via anonymized IDs
  - (Optional) Explore explicit text-to-table linking
- **Deliverables:** Annotated CARDIO:DE++ corpus, guidelines, reports, splits
- **Lead:** Heidelberg



**See also:** [CARDIO:DE Corpus Publication \(Richter-Pechanski et al., 2023\)](#)

# WP5: Advanced AI Model Development

- **Months:** 19–23
- **Objective:** Develop and evaluate advanced AI models
- **Key Tasks:**
  - Fine-tune LLMs for synthetic data
  - Develop data-to-text & text-to-data models
  - Patient phenotyping with multi-modal data
- **Deliverables:** Synthetic letters, evaluated models, phenotyping reports
- **Lead:** Heidelberg



**Further reading:** Med-PaLM: Towards Expert-Level Medical Question Answering with Large Language Models

# WP6: Dissemination, Publication & Sustainability

- **Months:** 19–24
- **Objective:** Maximize impact and ensure sustainability
- **Key Tasks:**
  - Public release of CARDIO:DE++ corpus
  - Documentation and usage examples
  - Peer-reviewed publications and workshops
  - Long-term maintenance plan
- **Deliverables:** Released corpus, documentation, publications, sustainability plan
- **Lead:** Heidelberg



**Open Science & FAIR Data:** FAIR Data Principles

# Timeline working packages



## Legend:

**WP1** Ethics, Legal Framework & Site Onboarding

**WP2** Data Acquisition

**WP3** De-identification

**WP4** Annotation

**WP5** Advanced AI Model Development

**WP6** Dissemination, Publication & Sustainability

# Summary: Expected Impact

First large, multi-modal, parallel German cardiovascular corpus





# Thank you!

